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Silurian paleogeography of Armorica: New paleomagnetic data from central Bohemia

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Abstract. Due to the lack of reliable paleomagnetic data from Armorica for mid-Ordovician to Late Devonian times, the drift history for this tectonic element is poorly understood. In order to help rectify this problem, a paleomagnetic study of Upper Silurian volcanic, intrusive and sedimentary rocks from the Barrandian Basin of the Central Bohemian Massif (Czech Republic) has been carried out. This basin comprises a sequence of unmetamorphosed lower Paleozoic rocks, the folding of which has an uppermost age limit of Early Carboniferous. Stepwise thermal demagnetization of the samples collected (182 samples from 21 sites), and principal component analysis of both great circle and stable endpoint data reveal a total of three directions of magnetization. The first (labelled A) is in the direction of the present-day local Earth's magnetic field (D/I=360°/67°) and is thus interpreted as being of recent origin. The other two directions, termed B and C, comprise the intermediate and high blocking temperature directions respectively. The overall in situ mean direction obtained for B, identified between 150° and 320°C, is 195°/-8° (D/I), k=18.5, α_{95}=13.2° (eight sites). It fails the fold test and yields an in situ paleo-south pole position of 42°S, 354°E. This coincides with the Late Carboniferous segment of the apparent polar wander (APW) path for stable Europe and is considered to be a secondary overprint of this age. The third magnetization (termed C) is identified between 320° and 600°C, gives an overall mean direction of 208°/-40°, k=96.8, α_{95}=5.7° (eight sites), after bedding correction. It passes fold and reversal tests on both local and regional levels and is interpreted as representative of the Late Silurian paleofield direction in the Bohemian Massif; it yields a paleo-pole position of 55°S, 325°E. Assuming a normal polarity, the inclination value obtained translates into paleolatitudes of 23°S for this part of Armorica and implies a position adjacent to the southern margin of Baltica. This therefore indicates closure of the intervening ocean by Late Silurian times. However, adopting this polarity also implies large-scale (up to 140°) anticlockwise rotations of the Bohemian Massif prior to deformation in the Early Carboniferous.

Introduction

Paleogeographic reconstructions for the circum Iapetan tectonic elements during Silurian times are problematic. Although the paleopositions of Laurentia and Baltica during the Early Paleozoic are now fairly well established from paleomagnetic data [*Van der Voo*, 1988; *Torsvik et al.*, 1990a; *Perroud et al.*, 1992] the Silurian location of Armorica, which originated from the northern margin of Gondwana in earliest Paleozoic times, remains unclear.

Recent paleomagnetic data from Avalonia, comprising coastal New England, Nova Scotia, Newfoundland, SE Ireland, southern Britain and the Ardennes, show that it rifted from Gondwana in Early Ordovician times. In conjunction with paleobiogeographical evidence, these data indicate closure of the Tornquist sea between Avalonia and Baltica prior to the Wenlock, and similarly a pre-Wenlock closure of the British sector of the Iapetus Ocean [*Torsvik et al.*, 1990a; *Channell et al.*, 1992; *Torsvik et al.*, 1993].

The picture for Armorica, however, which comprises the Hercynian Massifs within continental Europe, is less clear. Faunal provinciality and lithofacies indicators reflect a northward movement of Armorica and a change from Gondwana to Baltic affinities throughout the early Paleozoic. The presence of typical *Hirnantia* fauna and associated glacio-marine deposits in the Bohemian Massif and Iberia indicate continued high to intermediate paleolatitudes during the latest Ordovician [*Brenchley et al.*, 1991; *Owen et al.*, 1991]. Similarly, although the Silurian is characterized as a period of low provinciality for marine invertebrates [*Cocks and Fortey*, 1990; *Tuckey*, 1990], the occurrence of *Clarkeia* in northern Africa [*Cocks and Fortey*, 1990] and of diverse faunas, which include *Clarkeia*, in Early Silurian

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times in Armorica is indicative of a cold climate. Clastic sedimentation persisted in Armorica throughout Early Silurian times, but by the mid-Silurian the first carbonates had appeared. By the Pridoli there was substantial carbonate sedimentation which continued throughout the Devonian. This is similar to the sedimentation pattern elsewhere in Europe, where there was substantial reef building in the Pridoli [*Boucot*, 1990], but is in contrast to the record in Gondwana where clastic sedimentation persisted throughout Silurian times, with carbonates appearing first in the Devonian. Also, by Late Silurian times there was substantial mixing of faunas between North America, Baltica, Siberia, and the Bohemian Massif [*Tuckey*, 1990] reflecting the change in faunal affinities of Armorica from Gondwana to the northern continents.

This general pattern is also reflected in the existing paleomagnetic data set, although the lack of data from Armorica has caused the most recent apparent polar wander (APW) paths for this element to have been constructed by extrapolation between the published Ordovician and Devonian paleopoles [*Torsvik et al.*, 1990*b*]. Nevertheless, these data show a change in affinity of Armorica from Gondwana in Ordovician times [*Duff* 1979; *Perroud and Bonhommet*, 1981; *Perroud et al.*, 1986; *Tait et al.*, 1993] to the northern continents by Devonian times [*Van der Voo*, 1983]. However, the extent of the ocean between Armorica and Baltica and the relationship between Armorica and Gondwana during the Silurian have not yet been clearly established.

The only paleomagnetic data available for the Silurian of Armorica are derived from the San Pedro red beds [*Perroud and Bonhommet*, 1984], and the Almaden volcanic rocks [*Perroud et al.*, 1991] in Spain. The Almaden results pass a fold test and are interpreted by the original authors as being primary in origin and indicating paleolatitudes of 20°N. However, on the basis of paleomagnetic results from both underlying Ordovician volcanic rocks and overlying Devonian rocks from the axis of the Almaden syncline, *Parés and Van der Voo* [1992] have proposed a reinterpretation. They consider the directions obtained to be secondary in origin and interpret them as being of normal polarity and Carboniferous in age.

The San Pedro results, which indicate paleolatitudes of 18°S, also pass a fold test indicating that the magnetization was acquired prior to the Late Carboniferous folding. However, a pre-folding or syn-folding remagnetization of the San Pedro red beds, which come from a structurally very complex area, cannot be ruled out as discussed by *Van der Voo* [1993].

In order to constrain the position of the Bohemian Massif, and hence the eastern extremity of Armorica, in latest Silurian times, a detailed study of unmetamorphosed upper Wenlock/lower Ludlow, upper Ludlow, Pridoli, and lowermost Devonian carbonate and volcanic rocks from the Barrandian Basin in central Bohemia has been carried out.

Geology and Sampling

The Barrandian Basin forms a syncline within the gently folded Cadomian basement of central Bohemia. The basement has undergone low grade metamorphism to lower greenschist facies, and the basin fill comprises a sequence of unmetamorphosed Lower Cambrian to Givetian rocks, unconformably overlain by Westphalian age sedimentary rocks. The Barrandian sequence comprises Lower Cambrian conglomerates overlain by Ordovician clastic rocks of low diversity with Hirnantian age glacio-marine horizons, in turn overlain by Silurian and Devonian age shales and carbonates, with the first carbonate rocks appearing in mid Silurian times. Volcanic activity is represented predominantly by effusive basalts which persisted from Late Wenlock to the Ludlovian, with associated subsurface sill emplacement [*Havlíček*, 1981].

From the structural and stratigraphic relationships, and by comparison with surrounding areas in the Tepla-Barrandian where deformation was initiated in late Givetian times with a thermal event dated at 380 Ma from K-Ar studies [*Müller-Sohnius et al.*, 1987], deformation in the Barrandian Basin is thought to be Late Devonian to Early Carboniferous in age [*Franke*, 1990]. There is also evidence of synsedimentary deformation within the basin. Studies of the thickness and facies development by *Havlíček* [1981] have shown that much of the faulting and related folding resulted from a long, essentially synsedimentary process that continued throughout the Ordovician and Silurian and culminated after the close of sedimentation in the Givetian.

Major parts of the Barrandian Basin are located within a national park or have been declared as sites of special scientific interest. Thus only a limited number of sites could be sampled for paleomagnetic research. This is particularly true for the Siluro-Devonian boundary of which the strato-type locality and auxiliary sections are located along the Suchomasty and Berounka rivers, respectively (Figure 1). A total of 21 sites from volcanic, intrusive, and sedimentary rocks from both limbs of the basin were sampled. The oldest rocks sampled (sites MOT1-MOT4, Figure 1*a*) were collected from basaltic sills intruded into lower and upper Wenlockian sedimentary rocks. They represent shallow subsurface members of the Svaty Jan effusive-explosive volcanic complex [*Chlupáč et al.*, 1992] and have been dated as *circa* 420 Ma using K-Ar methods [*Fiala*, 1974]. Thus the structural correction applied to paleomagnetic results obtained from these rocks is that of the adjacent sediments. Sites KRA1-KRA3, LIT1-LIT3 and KRO1-KRO6 are located along the Suchomasty River (Figure 1*b*) and are of Ludlovian to Pridolian age, and sites HOS, KLO1-KLO3 and KST are located along the Berounka River (Figure 1*b*). Sites KLO1-KLO3 comprise pink and grey limestones and are situated at the auxiliary type section for the Siluro-Devonian boundary. For sites yielding interpretable paleomagnetic results the structural attitude, and lithologies and results are listed in Table 1 and Table 2, respectively. With the exception of site KLO2, the samples were collected from beds of uniform dip within site, but intersite variation allows for a fold test on a regional level. For KLO2 the samples were collected from around the limbs of an axial planar fold, thus also allowing for a fold test on the local level. The structural attitude of the volcanic and intrusive rocks were determined from immediately overlying sedimentary sequences.

Laboratory Procedures

The field-drilled cores were cut into standard 2.2 x 2.5 cm specimens, and magnetizations were measured using a two-axis "Remanomètre à squid RS 01" (LETI, Grenoble) cryogenic magnetometer in Munich or a 2G cryogenic

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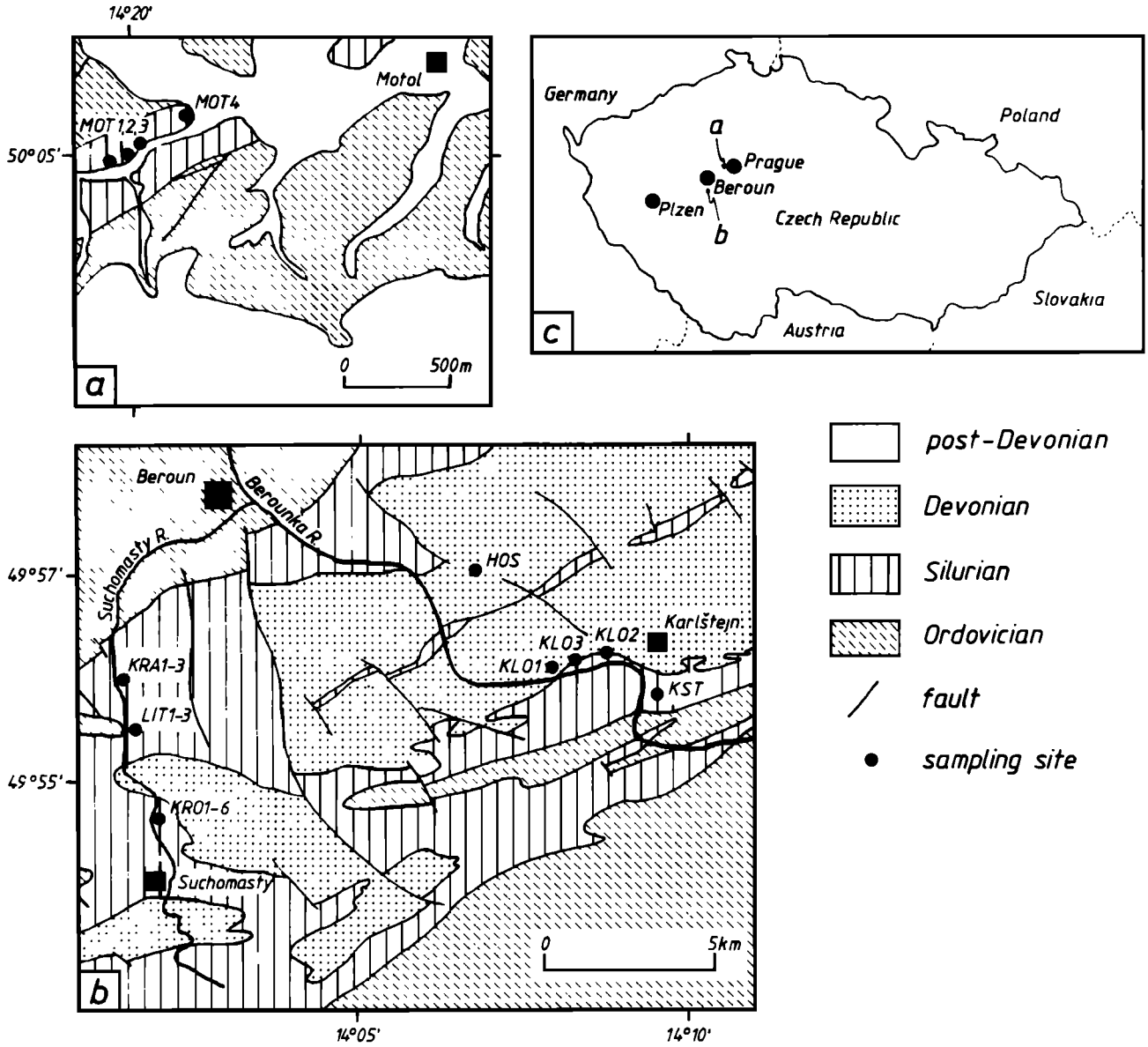


Figure 1. Geological sketch maps of the study areas within the Barrandian Basin, and the geographical location in eastern Europe.

magnetometer at the Swiss Federal Institute of Technology (ETH) Zurich. Samples were demagnetized using stepwise thermal or alternating field (AF) techniques, with measurement of the susceptibility of magnetization using a KLY2 Kappameter after each heating step.

The demagnetization results were analyzed using orthogonal vector plots [*Zijderveld*, 1967] and stereographic projection. Linear (planar) demagnetization trajectories were defined by at least three (four) successive data points which were selected by eye, and directions of the linear segments and poles to the planes were calculated using the principal component analysis of *Kirschvink* [1980]. Where overlapping coercivity or blocking temperature (T_B) spectra prevented complete separation of the components, great-circle analysis using the method of *McFadden and McElhinny* [1988] was carried out on the planar demagnetization trajectories to establish the direction of the underlying vector. This involves an iterative procedure whereby the maximum likelihood estimate of the endpoint direction within acceptable sector constraints for each great circle can be determined. The great circles are combined with stable endpoint directions, resulting in determination of the best estimate of the mean direction, with the statistics calculated in all cases using *Fisher* [1953] statistics.

Paleomagnetic Results

The natural remanent magnetization (NRM) intensities varied from 0.04 to 0.8 mA m^{-1} for the sediments and from 3 to 284 mA m^{-1} for the igneous rocks. Most samples measured show multivectorial behavior during thermal demagnetization. The sedimentary rocks in particular tend to undergo mineralogical alteration during thermal treatment, and so AF demagnetization was attempted. However,

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this method failed in most cases to reveal the full complexity of the demagnetization trajectories and so the results presented in this paper are based almost exclusively on thermal treatment. In this section, all the demagnetization results are presented in geographic coordinates unless otherwise stated.

During thermal demagnetization up to three directions of magnetization were identified; these are called A, B, and C and represent the low, intermediate, and high laboratory unblocking temperature components, respectively. Magnetization A is removed below 300°C and can be identified in samples from most sites. It is generally parallel to the local present-day Earth's magnetic field and is interpreted as being of recent origin. Most samples then show removal of an intermediate to high T_B magnetization (B direction). This is a shallow southerly direction which often persists until total demagnetization. Of the 21 sites studied, 10 revealed the presence of a third C direction of magnetization (MOT1-MOT4, KLO2, KST, KRO6 and KRA1-KRA3, Table 2), generally after removal of A and/or B. Of the remaining 11 sites, four yielded reliable site mean directions only for direction B (KLO1, KLO3, LIT1, and LIT3, Table 2); one was too weakly magnetized for accurate measurement (HOS), and samples from the other six sites were either dominated by direction A or there was no within site consistency of results (KRO1-KRO5 and LIT2). Thus results from sites HOS, KRO1-KRO5 and LIT2 are not listed in Table 2 and are not discussed further.

The volcanic and intrusive rocks (MOT1-MOT4, KST, KRA1-KRA3, LIT1, and LIT3) tend to show fairly simple behavior during thermal demagnetization (Figures 2*a* and 2*b*) with removal of a steep direction of magnetization between 20 and 250°C, followed by removal of B at intermediate unblocking temperatures. This magnetization is not present in all sites, but in some samples (e.g. sites LIT1 and LIT3) it persisted until complete demagnetization. After removal of B and/or A, continued demagnetization generally yields a well-defined C direction of mixed polarity, often with fairly narrow T_B spectra (Figure 2*b*). The maximum T_B in these rocks is between 550° and 600°C suggesting magnetite as the remanence carrier, which is supported by isothermal remanent magnetization (IRM) and thermomagnetic experiments.

Ludlovian tuffite beds from site KRO6, and the Pridoli detrital carbonates from site KLO2, (examples of which are shown in Figures 3*a* and 3*b*, respectively) also reveal the presence of a third, higher T_B direction which is of opposing polarity between these two sites. In the tuffites, the isolation of directions of A and B at lower T_B's is not always clear. However, above temperatures of 350°C, direction C can be identified with intermediate negative inclination values (Figure 3*a*). The magnetic carrier is thought to be magnetite, although there is evidence for a contribution by hematite (Figure 3*a*).

Table 1. Structural Attitude of Beds Sampled

		Fold at Site KLO2	
Site	Strike/Dip(°)	Sample	Strike/Dip(°)
KLO3	230/55	KLO2(1-4)	235/70
KLO1	242/55	KLO2(5-6)	234/53
KST	246/43	KLO2(7-8)	051/33
KRO6	360/30	KLO2(9)	052/19
KRA1-KRA2	281/18	KLO2(10)	055/32
KRA3	251/16	KLO2(11-14)	053/26
LIT1	084/32		
LIT3	095/29		
MOT1-MOT4	275/21		

For the carbonate samples (Figure 3*b*), demagnetization characteristics are more complex. There is often a decay of up to 50% in the intensity of magnetization between 20° and 150°C, representing the removal of component A. The B direction is then removed at intermediate temperatures, between 200 and 320°C, during which demagnetization trajectories follow a great-circle path due the to presence of the third, higher T_B C magnetization. Stable endpoints are only reached above 320°C after removal of both A and B (Figure 3*b*). Intensity decay curves for these samples are characterised by an inflection between 300° and 350°C.

However, due to thermochemical alteration, stable endpoints are not always reached, and the magnetization directions become random with a marked increase in the susceptibility of magnetization above 350°C, typically accompanied by an increase in the intensity of magnetization. This scenario is illustrated in Figure 4*a* where the demagnetization trajectory continues to curve through the last measurement. Stereographic projection of the results obtained from successive demagnetization steps shows two distinct great-circle paths (Figure 4*b*). The first great circle described between 20° and 150°C results from removal of the A and B directions; the second, between 200° and 400°C, moves towards a north/northeasterly direction and results from combined removal of B and C. It is clear therefore that the endpoint must lie somewhere on the great circle after the last measurement. In sites such as this, the combined stable endpoint and great-circle analysis of *McFadden and McElhinny* [1988] was employed in order to determine the direction of the underlying component.

Curves obtained from IRM experiments show a rapid increase in intensity of magnetization up to 0.4 T, by which field at least 95% of the magnetization is acquired. This is followed by a small increase up to 1.5 T, without reaching saturation. Taking all the characteristics observed into account, we interpret the low T_B (< 150°C) direction (A) as being carried by goethite (with coercivity > 1.5 T), and the secondary B direction of magnetization as carried by pyrrhotite which has coercivities up to 0.4 T and a maximum T_B of 350°C. The marked increase in susceptibility of magnetization noted at higher temperatures is typical behavior for pyrrhotite during thermal treatment. Well-defined stable endpoints of the underlying C direction are only reached at temperatures between 340°C and 450°C, i.e., above the maximum T_B of pyrrhotite. It is thought that this magnetization vector is carried by magnetite, the presence of which has been verified in thermomagnetic experiments.

Interpretation

The B magnetization can be reliably identified in eight of the 21 sites and the in situ sample directions are shown in Figure 5*a*. Combining the site mean data listed in Table 2, yields an overall in situ mean direction of 195°/-8°, $k = 18.5$, $\alpha_{95} = 13.2°$. Bedding correction results in a significant decrease of k ($k_2/k_1 = 0.405$ where k_1 and k_2 are the precision paramters of *Fisher* [1953] before and after

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Table 2a. Site Mean Directions and Overall Means for Component B

Site	Lithology	Age	*N/n*	In Situ Dec/Inc	*k*	α_{95}
KLO1	pink limestone	Ged	12/10	192/−22	48.5	5.8
KLO2	grey limestone	Prid	14/8	199/+06	8.8	40.2
KLO3	grey limestone	Prid	7/4	197/−29	38.2	15.1
KST	basalt	P/L	9/0	−/−	–	–
KRO6	tuffite	P/L	8/0	−/−	–	–
KRA1	basalt	P/L	7/6	186/−02	20.5	15.2
KRA2	basalt	P/L	9/4	219/+12	11.3	28.5
KRA3	basalt	P/L	4/0	−/−	–	–
LIT1	basalt	P/L	4/4	183/−04	18.6	21.9
LIT3	basalt	P/L	5/5	184/+00	42.1	11.9
MOT1†	basalt	U Wen	7/2	217/−04	–	–
MOT2†	basalt	U Wen	7/2	195/−05	–	–
MOT3	basalt	U Wen	7/3	203/−21	195.0	8.8
MOT4	basalt	U Wen	6/0	−/−	–	–
Mean			eight sites	195/−08	18.5	13.2
Paleopole				42°S, 354°E		

Dec, declination; Inc, inclination. Ged, Prid, P/L, and U Wen represent Pridoli, Pridoli/Ludlow and upper Wenlock, respectively. *N*, the number of samples measured; *n*, number of sample directions used in the calculation of the site mean directions. Also listed are the precision parameter *k* of *Fisher* [1953] and the half angle for the cone of 95% confidence [*Fisher*, 1953].

† sites not used in calculation of the overall mean.

Table 2b. Site Mean Directions and Overall Means for Component C

Site	*N/n*	In Situ Dec/Inc	*k*	α_{95}	Bedding Corrected Dec/Inc	*k*	α_{95}
KLO1	15/0	−/−	–	–	−/−	–	–
KLO2	14/10(4)	070/+36	4.9	24.3	045/+49	40.5	7.8
KLO3	7/0	−/−	–	–	−/−	–	–
KST	9/9	255/−48	47.9	7.5	212/−37	47.9	7.5
KRO6	8/7	179/−52	77.1	6.9	212/−43	77.1	6.9
KRA1†	7/2	062/+68	–	–	040/+54	–	–
KRA2	11/7	211/−58*	68.3	7.4	205/−41*	68.3	7.4
KRA3†	4/2	226/−69*	–	–	199/−59*	–	–
LIT1	4/0	−/−	–	–	−/−	–	–
LIT3	5/0	−/−	–	–	−/−	–	–
MOT1	7/5	215/−49	65.0	9.6	207/−30	65.0	9.6
MOT2	7/7	203/−56	41.1	8.9	197/−36	41.1	8.9
MOT3	7/6	218/−61	83.2	7.4	206/−42	83.2	7.4
MOT4	6/5	219/−62	27.4	15.0	206/−43	27.4	15.0
Mean	eight sites	220/−55	21.8	12.1	208/−40	96.8	5.7
Paleopole					55°S, 325°E		

Notation as in Table 2*a*. Number in brackets indicates number of great circles used to compute the site mean direction.

* sites in which directions of mixed polarity were identified.

† sites not used in calculation of the overall mean.

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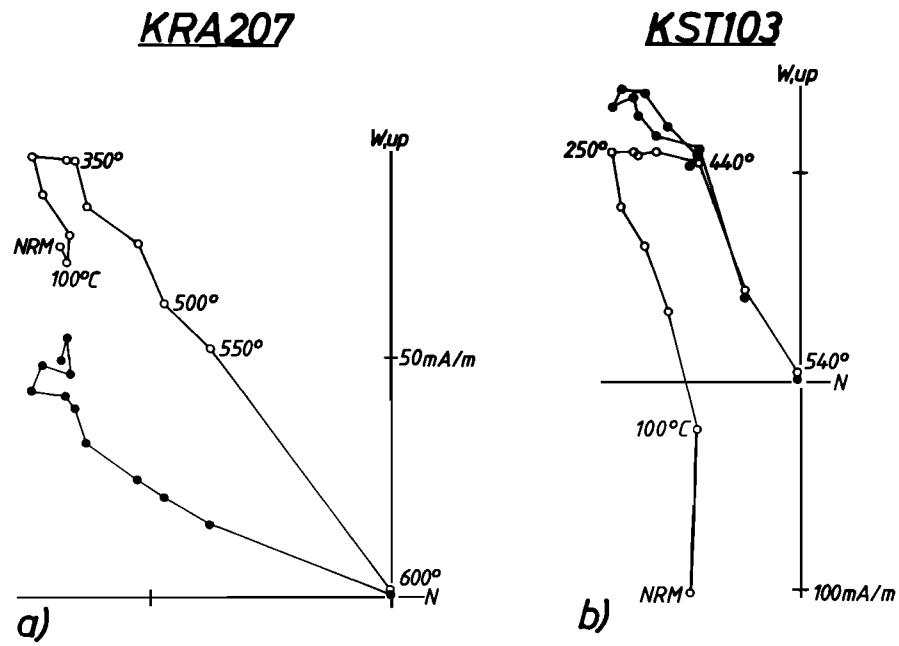


Figure 2. Thermal demagnetization results from volcanic rocks. All directions are plotted in geographical coordinates, solid (open) symbols represent the horizontal (vertical) component. Both these examples show removal of A between 20° and (*a*) 280°C or (*b*) 250°C, followed by removal of B at intermediate temperatures. At temperatures higher than 350°- 440°C, C is identified as an intermediate negative and southwestward direction of magnetization.

bedding correction, respectively). Component B therefore fails the fold test, demonstrating acquisition after deformation and is considered to be of secondary origin.

The uppermost age limit for folding of these rocks is Early Carboniferous, during the Hercynian orogeny. By this time it is clear from both geological and paleomagnetic evidence [*Van der Voo*, 1983] that the Bohemian Massif and Armorica were part of the Old Red Continent. The paleopole position for component B at 42°S, 354°E clearly coincides with the Late Carboniferous sector of the European APW path, as given by *Van der Voo* [1993] and is therefore considered to represent an overprinting event of this age.

Where directly observed, magnetization C can generally be isolated as linear segments directed toward the origin in orthogonal projection. Otherwise, great-circle and stable endpoint data were combined using the procedure of *McFadden and McElhinny* [1988]. Using this technique for site KLO2, a best fit in situ site mean solution of 070°/36°, k=4.9, α_{95}=24.3°, and bedding corrected solution of 045°/49°, k=40.5, α_{95}=7.8° (10 samples), is obtained. There is a marked improvement in the clustering of observed stable endpoints and great-circle intersections after bedding correction (Figures 6*a* and 6*b*), with a maximum value of k at 95% unfolding (Figure 6*c*). This improvement is significant at the 99% level of confidence using the fold test of *McElhinny* [1964]. The sample populations were also subjected to the test of *McFadden and Lowes* [1981] to determine for both the in situ and bedding corrected values whether they could have been drawn from populations sharing a common mean. The null hypothesis can be rejected in situ, but there is more than 99% likelihood that the bedding corrected directions are drawn from a Fisherian distribution with a common mean, indicating a prefolding age of magnetization for this site.

Although the rock ages range from upper Wenlock to Pridoli, there is no significant difference in the site mean directions with respect to age. Therefore all the site mean data with n>5 (Table 2) were combined, resulting in an overall mean direction of 220°/-55°, k=21.8, α_{95}=12.1° in situ, and 208°/-40°, k=96.8, α_{95}=5.7° (eight sites) after bedding correction. The improvement in site mean grouping (compare Figures 7*c* and 7*d*) reaches a maximum after 95% unfolding (Figure 7*e*) which is significant at the 99% level of confidence using the fold test of *McElhinny* [1964].

The reversal test of *McFadden and McElhinny* [1990] was applied to the bedding-corrected site mean directions (Table 2). The supplement of the antipode is less than the critical angle of 12° at the 95% confidence level; thus the directions pass the reversal test with classification C. This observation and the fact that the fold test is passed at both the local and regional levels support the interpretation that the magnetization is prefolding and closely linked to the age of deposition. A paleopole position of 55°S, 325°E is obtained from these results and is considered to be representative for Armorica from late Wenlock through to late Pridoli times (approximately 425-408 Ma).

Discussion and Conclusions

In the overall context of tectonic evolution of the Iapetus bordering continents, it is now evident that after development of the Tornquist sea in Ordovician times, drifting of Avalonia from the northern margin of Gondwana resulted in a pre-Wenlock closure of the Tornquist between Avalonia and Baltica [*Torsvik et al.*, 1993]. By using the new paleomagnetic data for Armorica presented in this paper, it is now possible to constrain the Paleozoic evolution of the ocean between Baltica and Armorica. Assuming a normal

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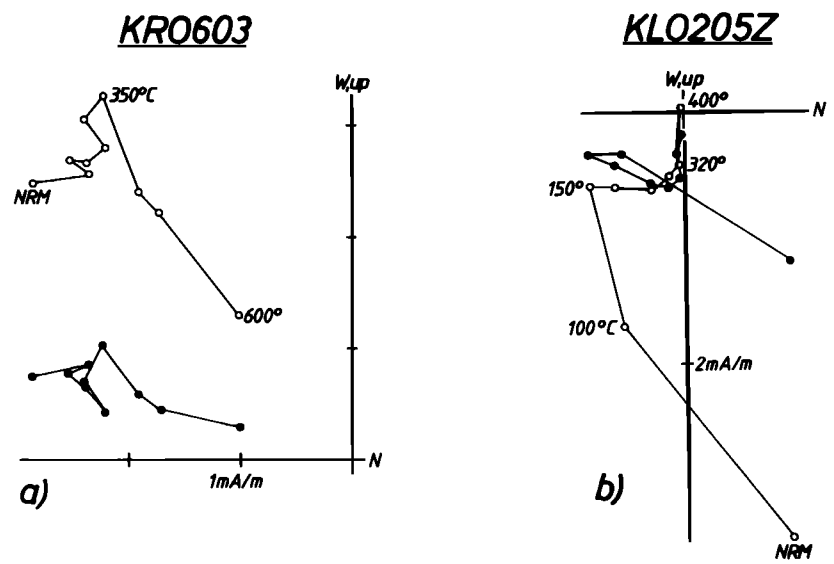


Figure 3. In situ thermal demagnetization characteristics of (*a*) the tuffites revealing a south-southwest negative C magnetization direction above 350°C and (*b*) the carbonates in which all three directions of magnetization can be clearly identified. Notation is as for Figure 2.

polarity for the C magnetization places the Bohemian Massif at 23°S from late Wenlock to late Pridoli times. Comparing this with the well-constrained Upper Silurian-Lower Devonian position of Baltica [*Douglass*, 1988], it is clear that Armorica was adjacent to its southern margin during this time (Figure 8), thus indicating latitudinal closure of the intervening ocean by the late Wenlock. However, adopting this polarity option requires considerable rotation of the Bohemian Massif before acquisition of the B magnetization, and consolidation into the present-day configuration of Hercynian Europe (Figure 8). Nevertheless, it is preferred to the reversed polarity option which would place the Bohemian Massif at paleolatitudes to the north of Scandinavia, which is considered unlikely. Our results therefore are interpreted as representing the onset of collision between Armorica and Baltica rather than final suturing.

The position of Gondwana shown in Figure 8 is based on the arguments of *Van der Voo* [1993]. Whether or not this reconstruction for Gondwana is valid depends crucially on the interpretation of the paleomagnetic data set. An alterna-

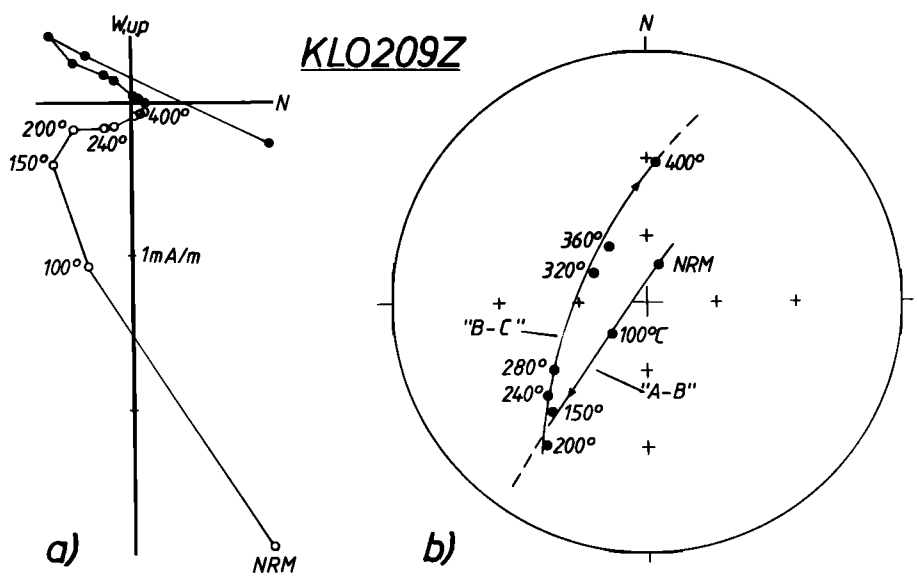


Figure 4. (*a*) Orthogonal projection of thermal demagnetization results, notation is as for Figure 2, and (*b*) corresponding equal area stereographic projection of a carbonate sample, where solid (open) symbols represent lower (upper) hemisphere projections, in which stable endpoints are not reached. In Figure 4(*b*) two great-circle paths can be clearly identified, the first between 20° and 150°C, resulting from removal of A and B directions of magnetization, and the second from 200°- 400°C, resulting from removal of B and C directions.

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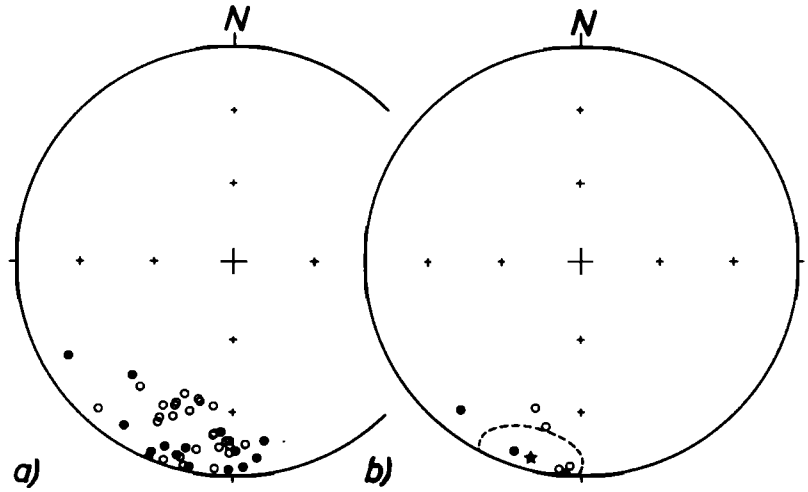


Figure 5. Equal area projection of the (*a*) in situ sample and (*b*) site mean directions for B. The star in Figure 5(*b*) represents the overall mean direction with its cone of 95% confidence. Notation is as for Figure 4.

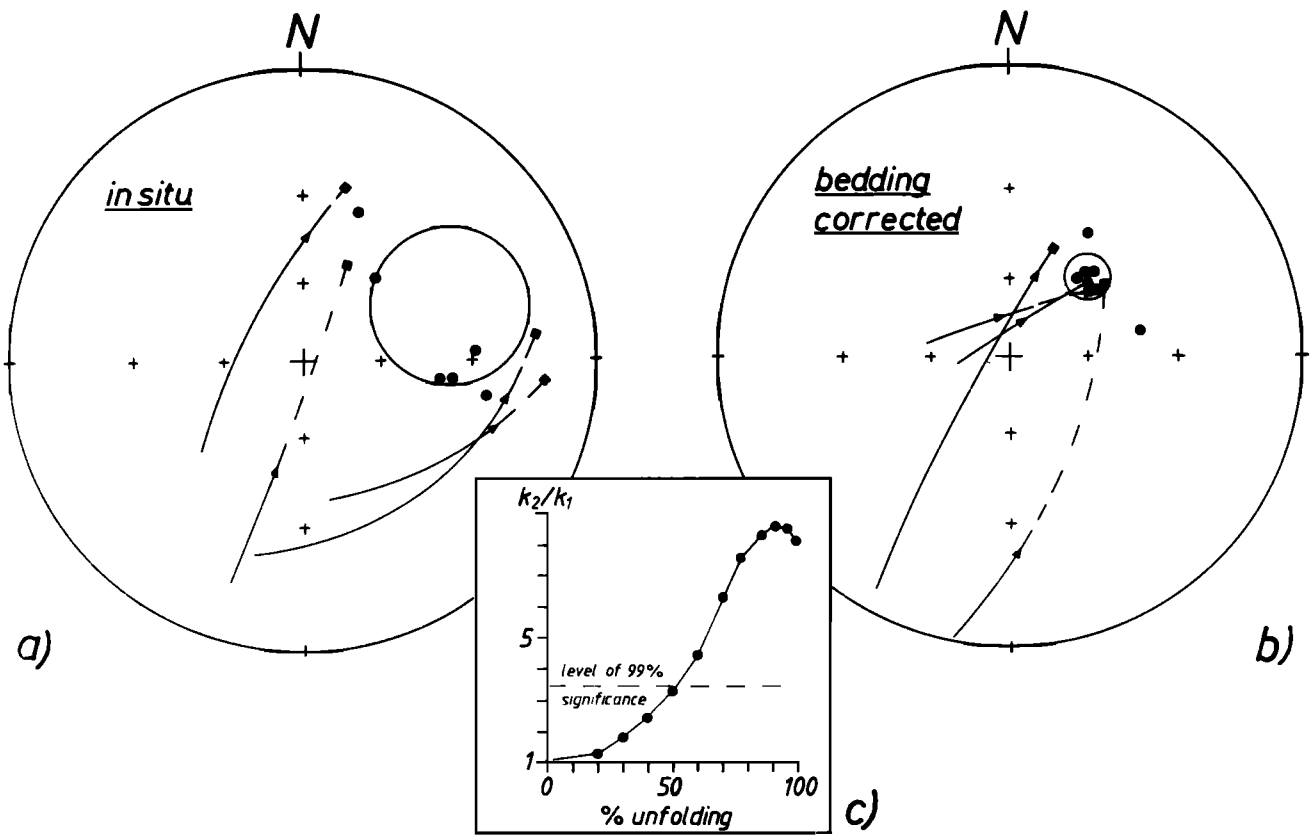


Figure 6. Equal angle projection of the (*a*) in situ and (*b*) bedding corrected great circles and stable endpoints identified for the carbonate rocks of site KLO2. The square on each great circle represents the best fit solution for the direction of the endpoint within the given sector constraints. Also plotted is the cone of 95% confidence for the site mean direction. (*c*) shows the k_2/k_1 ratio versus percentage unfolding.

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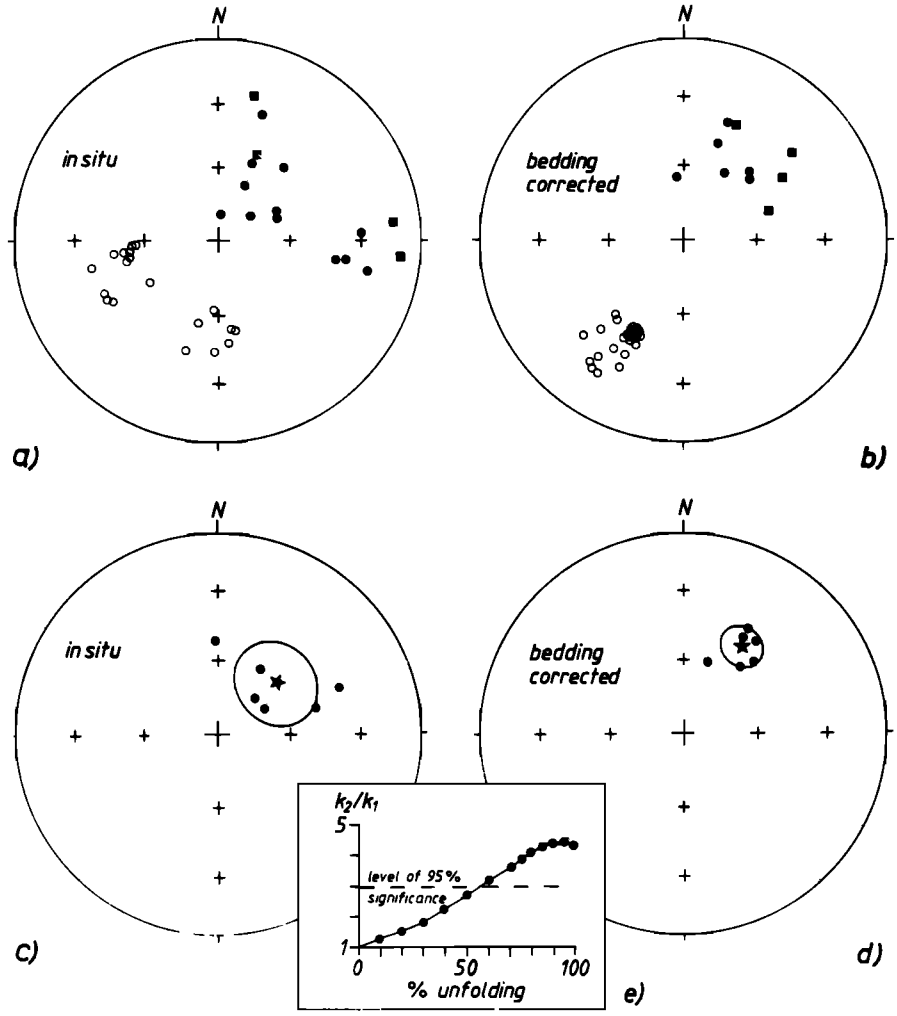


Figure 7. Equal area projections of the (*a*) in situ, and (*b*) bedding corrected sample directions for C. Square symbols represent the directions obtained from great-circle data. (*c* and *d*) Site mean directions. Those with negative inclination values have been inverted. Also shown is the cone of 95% confidence and the overall mean direction both before and after bedding correction. (*e*) The k_2/k_1 ratio *vs* percentage unfolding.

tive interpretation has been suggested by *Bachtadse and Briden* [1991], who argued that the northern margin of Gondwana remained at high paleolatitudes until Devonian times.

If the paleomagnetic data presented in this paper are used, the Bohemian Massif is positioned at higher paleolatitudes than that expected from the Siluro-Devonian age paleopole position obtained from San Pedro, northern Spain [*Perroud and Bonhommet*, 1984]. This may imply either a lack of coherence between these two elements of Armorica (Iberia and Bohemia), as proposed by *Ziegler* [1990], or a younger age of magnetization in the San Pedro, as suggested by *Van der Voo* [1993]. The former hypothesis cannot be ruled out because the coherency of Armorica during Paleozoic times is still far from clear. Although Armorica is generally referred to as a single microplate in the Iapetan system, it may comprise a mosaic of semiautonomous tectonic elements [*Ziegler*, 1990].

Paleomagnetic data from Lower Ordovician rocks in the Armorican, Iberian, and Bohemian massifs of Armorica are in agreement and indicate high peripolar paleolatitudes [*Duff*, 1979; *Perroud and Bonhommet*, 1981; *Perroud et al.*, 1986; *Tait et al.*, 1993]. However, the picture for Late Ordovician and Silurian times and the relationship between Gondwana and Armorica are more enigmatic. This is partly due to the lack of reliable data for this time period for Armorica, but it is also due to apparent contradictions within the paleomagnetic data set for Gondwana. If the reconstruction of *Van der Voo* [1993] shown in Figure 8, placing the northern margin of Gondwana at low paleolatitudes in Silurian times, is adopted, then this suggests there was no separation between Armorica and Gondwana during lower Paleozoic times. Alternatively, if the arguments of *Bachtadse and Briden* [1991] for a position of Gondwana at high latitudes during this time interval are accepted, then a wide separation between Gondwana and Armorica is implied.

The new primary Silurian data demonstrate that prior to consolidation of Hercynian Europe and partial remagnetization during Carboniferous times, the Bohemian Massif

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Figure 8. Late Wenlock - late Pridoli paleogeographic reconstruction of the main Paleozoic tectonic elements in the Iapetus/Tornquist systems. Armorica is positioned according to data from the Barrandian Basin (this study), Baltica according to the Ringerike pole of *Douglass* [1988], Avalonia in its Wenlock position after *Torsvik et al.* [1993], Laurentia according to *Van der Voo* [1988], and Gondwana after *Van der Voo* [1992]. The location of the Bohemian Massif is marked by a star.

underwent major anticlockwise rotation (to the order of 140°). The resulting paleopole position cannot therefore be used to construct an APW path for Armorica. Nevertheless, the inclination value obtained indicates that the Bohemian Massif was adjacent to the southern margin of Baltica/Avalonia, with closure of the intervening ocean by the Late Silurian.

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